

Lova Chakravarthi Dammu

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Professional Summary

Blockchain Engineer with 3+ years of experience designing and developing production-grade smart contracts, DeFi protocols, RWA tokenization systems, DAO governance, cross-chain protocols, and enterprise blockchain infrastructure on EVM-compatible networks. Experienced in ERC-3643 / Tokeny T-REX compliant asset tokenization, OnchainID-based identity and KYC/AML controls, DAO treasury and governance systems, staking and locking mechanisms, atomic swaps and HTLC-based settlement, blockchain bridges, oracle integration, and permissioned Hyperledger Besu networks. Strong background in Solidity, EVM internals, gas optimization, smart contract security, validator infrastructure, HD wallets, cryptographic signing, and on-chain/off-chain protocol architecture.

Core Competencies

Blockchain Protocol Engineering: Solidity, EVM, Smart Contracts, Protocol Architecture, DeFi, RWA, Tokenization, Tokenomics, DAO Governance, Cross-Chain Protocols, Permissioned Blockchain

DeFi & Financial Protocols: Staking, Yield Management, Liquidity Management, Treasury Management, Vaults, Escrow, Lock-to-Earn, Token Locking, Reward Distribution, Vesting, Atomic Swaps, HTLC, Cross-Chain Settlement

RWA & Compliance: ERC-3643, Tokeny T-REX, OnchainID, KYC/AML, KYA, Permissioned Tokenization, Asset Classification, RBAC, Whitelisting, Blacklisting, Compliance Gates, Entity Registry

DAO & Governance: Proposal Lifecycle, Voting, Vote Delegation, Lock-Weighted Voting, Governance Tokens, Treasury Governance, Multisig Governance, Veto Mechanisms, Emergency Governance

Cross-Chain: Blockchain Bridges, Cross-Chain Messaging, Relayers, Multisig Relayers, Cross-Chain Contract Calls, Atomic Swaps, HTLC, Nonce Management, Replay Protection, Chain Whitelisting, Emergency Pause

Security: Access Control, RBAC, Multi-Signature Security, ECDSA, Replay Protection, Nonce Management, Signature Verification, Smart Contract Auditing, Gas Optimization, CTFs, Key Management

Professional Experience

Solidity Blockchain Developer

Gaian Solutions

Aug 2024 – Present

Hyderabad, India

- Architected and developed RWA tokenization protocols using **ERC-3643** and the **Tokeny T-REX** framework for compliant on-chain representation of real-world assets.
- Integrated **OnchainID** with third-party KYC/AML services for wallet identity binding, investor eligibility, compliance enforcement, and token transfer authorization.
- Implemented **Role-Based Access Control (RBAC)**, wallet whitelisting/blacklisting, compliance gates, investor classification, and permissioned asset issuance.
- Designed modular **DAO governance infrastructure** supporting proposal creation, voting, execution, treasury management, vesting schedules, automated fund allocation, and gas-optimized reward distribution.
- Engineered **DeFi protocol components** including staking, asset locking, reward distribution, treasury/vault mechanisms, escrow flows, yield harvesting, and lock-based financial incentives.
- Designed **cross-chain protocol architecture** for RWA liquidity movement and settlement using blockchain bridge contracts, message relayers, cryptographic signatures, nonce-based replay protection, and emergency pause controls.
- Implemented **atomic swap and HTLC-based settlement flows** supporting create, certify, release, refund, and cross-chain asset settlement lifecycle.
- Integrated **Chainlink oracles** for off-chain asset valuation, price/data feeds, settlement conditions, and automated contract execution.
- Designed **permissioned Hyperledger Besu infrastructure** with node permissioning, private transactions, consortium governance, enterprise compliance controls, and EVM-compatible smart contracts.
- Implemented security controls including **multisig authorization, signature verification, replay protection, nonce management, role separation, emergency circuit breakers, treasury controls, and smart contract security reviews**.
- Designed on-chain/off-chain protocol architecture integrating event monitoring, relayers, registries, audit anchoring, evidence verification, and blockchain settlement services.

Blockchain Backend Developer

ULTRAPRO Blockchain Pvt. Ltd.

Dec 2023 – Jun 2024

Madurai, India

- Designed blockchain backend services for **BEP-20 deposits and withdrawals** across BNB Chain for BNB, UPRO, and USDT.
- Implemented **multi-signature verification, nonce management, event-driven transaction monitoring, transaction lifecycle tracking, and replay-attack protection**.
- Built deterministic **HD wallet infrastructure** using BIP-39 mnemonic generation and BIP-32/BIP-44 hierarchical key derivation across Ethereum, BNB Chain, and Polygon.
- Implemented **ECDSA transaction signing, encrypted mnemonic/key storage, wallet generation, trans-**

action submission, and blockchain state monitoring.

- Developed REST and WebSocket services integrating on-chain transaction events with real-time TradingView market feeds, analytics, and application-layer APIs.

Key Projects

Mobius Lattice – RWA / DeFi / DAO / Cross-Chain Protocol Architecture 2026

Solidity, Hyperledger Besu, EVM, Blockchain Bridge, DAO, RWA, HTLC, Multisig, Merkle Proofs

- Designed a multi-layer blockchain architecture consisting of a constitutional root, commons layers, human sovereignty layer, and independent permissioned EVM worlds.
- Designed **MobiusBridge** for bidirectional cross-chain communication using destination world IDs, signed payloads, relay services, monotonic nonces, chain whitelisting, and emergency direction-level pause controls.
- Designed **3-of-5 multisig relayer architecture** for cross-chain message delivery and signature validation.
- Designed RWA/token economy architecture involving **RASA assets, TAPAS governance, token locking, governance voting, treasury hooks, yield harvesting, and release/reconciliation flows.**
- Implemented the protocol lifecycle where locked assets generate governance rights, support **lock-weighted voting**, interact with treasury yield, and settle through deterministic release and accounting logic.
- Designed **MarketplaceRegistry, EvidenceRegistry, ParticipantRegistry, ConstitutionRegistry, EntityRegistry, TokenFactory, and WorldAnchorFactory** components for protocol-level asset and network management.
- Integrated **Merkle-root audit anchoring** and cross-chain evidence registration to provide auditable state references across independent permissioned networks.
- Designed **KYA/compliance gates, entity binding, investor classification, governance roles, root veto controls, treasury/endowment flows, and retained-rights mechanisms.**

CoreDAO Mainnet Validator Node 2025

AWS EC2, CoreDAO, Geth, Linux, systemd, JSON-RPC

- Deployed and operated a **CoreDAO mainnet full node** on AWS EC2 using a pruned snapshot synchronization workflow.
- Configured Linux/systemd service management for automatic restart and used UFW firewall policies for node hardening.
- Transitioned the node into validator infrastructure, configured consensus-related account settings, and validated blockchain health through JSON-RPC.
- Verified chain synchronization, peer connectivity, node health, and validator registration workflows.

Venom Blockchain DAO Protocol 2023

Solidity, TypeScript, Locklift, Venom Blockchain

- Delivered a production DAO protocol suite containing **Governance, Treasury, FounderNFT, DeedNFT, and CommissionDistribution** smart contracts.
- Built proposal creation, voting, execution, treasury release, and interval-based fund distribution lifecycle.
- Implemented **NFT-based governance eligibility** through dual DeedNFT and FounderNFT locking requirements.
- Integrated token swap logic with NFT-status-based eligibility controls and created TypeScript deployment, verification, and testing workflows using Locklift.

Technical Skills

- **Languages:** Solidity, JavaScript, TypeScript, Python, Go, Rust, Java
- **Blockchain:** Ethereum, EVM, BNB Chain, Polygon, CoreDAO, Venom, Hyperledger Besu, Hyperledger Fabric
- **Smart Contracts:** Solidity, OpenZeppelin, ERC-20, ERC-721, ERC-1155, ERC-3643, Upgradeable Contracts, Proxy Patterns, Token Factories, Vaults, Escrow
- **DeFi:** Staking, Yield Generation, Yield Harvesting, Liquidity Management, Treasury Management, Token Locking, Lock-to-Earn, Vesting, Reward Distribution, Atomic Swaps, HTLC
- **DAO Governance:** DAO Architecture, Proposal Lifecycle, Voting, Vote Delegation, Lock-Weighted Voting, Governance Tokens, Treasury Governance, Multisig Governance, Veto Mechanisms
- **RWA:** RWA Tokenization, ERC-3643, Tokeny T-REX, OnchainID, KYC/AML, KYA, Compliance Gates, Asset Classification, Permissioned Tokenization, Entity Registry
- **Cross-Chain:** Blockchain Bridges, Cross-Chain Messaging, Relayers, Multisig Relayers, HTLC, Atomic Swaps, Cross-Chain Settlement, Nonce Management, Replay Protection, Signature Verification, Chain Whitelisting
- **Security:** RBAC, Access Control, Multisig, ECDSA, Replay Protection, Nonce Management, Emergency Pause, Smart Contract Auditing, Gas Optimization, Key Management, CTFs
- **Oracles & Verification:** Chainlink, Oracle Integration, Data Attestation, Merkle Proofs, Merkle Roots, Evidence Registry, Audit Anchoring, Proof Verification
- **Enterprise Blockchain:** Hyperledger Besu, Hyperledger Fabric, Permissioned Networks, Private Transactions, Consortium Networks, Node Permissioning

- **Blockchain Infrastructure:** AWS EC2, Linux, Ubuntu, Docker, Geth, JSON-RPC, systemd, UFW, Validator Nodes, Full Nodes, Snapshot Sync, Node Monitoring
- **Development Tools:** Hardhat, Truffle, Ethers.js, Web3.js, Locklift, OpenZeppelin, IPFS
- **Backend & APIs:** Node.js, REST APIs, WebSocket, GraphQL, MongoDB, Blockchain Event Monitoring, Transaction Indexing
- **Frontend:** React, Next.js

Education

Bachelor of Technology

RGUKT AP-IIIT Nuzvid

2020 – 2024

Andhra Pradesh, India

- CGPA: 8.25 / 10
- Relevant Coursework: Distributed Systems, Computer Networks, Cryptography, Data Structures

Achievements

- **CoreDAO Mainnet Validator Node – 2026:** Deployed and operated a CoreDAO mainnet full node on AWS EC2 using a pruned snapshot synchronization workflow. Configured Linux/systemd service management for automatic restart and used UFW firewall policies for node hardening. Transitioned the node into validator infrastructure, configured consensus-related account settings, and validated blockchain health through JSON-RPC. Verified chain synchronization, peer connectivity, node health, and validator registration workflows.
- **GDSC WOW Hackathon – Google Developer Student Clubs, Kerala (2023):** Built a blockchain-based e-learning credential verification platform using decentralized records.
- **Smart Contract Security CTFs:** Solved Ethernaut challenges covering reentrancy, overflow, delegatecall, access control, and common smart contract vulnerabilities. Completed applied cryptography and vulnerability research challenges through picoCTF.

Certifications

- Full Stack Blockchain Developer – iNeuron (200-hour Bootcamp)
- Ethereum Blockchain Developer – Udemy